Project proposal: detecting ancestry from unsampled donor lineages

Status: proposal for written course approval; approval has not been recorded. November milestone: minimal presentable version and written approval by November 14. Final target: December 7 readiness; the assignment states December 8 submission.

Question and motivation

How do ArchIE and TRACE differ in their ability to recover DNA inherited from an unsampled donor population as admixture becomes older or rarer, and when their demographic or reference-panel assumptions are violated? A secondary question is how much TRACE's performance depends on reconstructing ancestral recombination graphs rather than observing their true simulated histories.

Human introgression studies motivate these approaches. Durvasula and Sankararaman (2020) used population-genetic summaries and ArchIE to map candidate archaic segments without a donor genome. Zhang and colleagues (2026) introduced TRACE, which uses features of inferred local genealogies. Their information requirements differ: ArchIE uses a comparison population and simulation-trained parameters; TRACE uses inferred genealogies. This project evaluates adapted software, not the claim that unsampled ancestry automatically identifies an extinct species.

Data and two approaches

The main dataset will consist of reproducible msprime simulations with donor migration records. Ten diploid target and ten diploid reference individuals will be simulated over independent 1-Mb regions. Neither detector receives donor samples or test ancestry labels. Migration records provide exact truth for evaluation, including mixed regions. Training, calibration and test seeds are separate. The controlled model is not a fitted mosquito population history.

ArchIE will retain its independently trained binomial classifier. TRACE will use SINGER-inferred ARGs; a separately labeled true-ARG analysis isolates reconstruction losses. Existing source adaptations and upstream commit identifiers are documented. An initial real software run has completed, but its short MCMC chain and one seed establish functionality rather than comparative accuracy.

Tests and expected interpretation

Primary scoring intersects predicted and true intervals across all callable bases. Six independent baseline validation regions select thresholds targeting FDR at most 5%, then thresholds are frozen. Test FDR may exceed that target under other histories. If no nonempty validation call set qualifies, that absence is a result, not a successful FDR guarantee.

Five to ten independent replicates will cover no introgression, different introgression ages, lower ancestry proportion, and an alternative ancestral history. The latter includes migration and is interpreted as an alternative-history call rate rather than a pure false-positive control. Optional stress tests add reference admixture and demographic mismatch. Confidence intervals resample independent simulation replicates, retaining pairing between methods.

We will compare recall, achieved FDR, false-called sequence in the null model, resource cost, and systematically selected localization errors. Null findings and uncertain comparisons are valid outcomes. A read-only local explorer will link performance plots to ancestry tracks; static figures and data remain reproducible.

Optional mosquito demonstration and deliverables

Ag1000G provides public phased mosquito genomes and metadata. An empirical pilot is conditional on defensible population parameters, QC/relatedness evidence, and ancestral allele orientation. Reference alleles will not be treated as ancestral alleles. Method agreement would be an empirical consistency check, not accuracy or proof of a new ghost lineage. Biological observations are exploratory only.

Final deliverables follow the assignment: about two introduction pages, 2-3 results pages excluding figures, and 1-2 discussion pages, with 12-point font and 1.5 spacing; three principal figures; and annotated executed code and environment records. Local experiments are capped at twelve elapsed hours with four workers.

References

Durvasula A, Sankararaman S. Recovering signals of ghost archaic introgression in African populations. Science Advances (2020). DOI: 10.1126/sciadv.aax5097.

Zhang Y et al. Recovering signatures of archaic hominin introgression using ancestral recombination graphs. Science (2026). DOI: 10.1126/science.aef8874.

Software: https://github.com/sriramlab/ArchIE ; https://github.com/YulinZhang9806/trace ; https://github.com/popgenmethods/SINGER . Data:

https://www.malariagen.net/data_package/ag1000g-phase3-hap/ .